Toward a Counterexample to Agrawal's Conjecture via Twin Smooth Integers
(working draft — v0.1)

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Abstract

Agrawal's conjecture asserts that for coprime n and prime r, the congruence $(X-1)^n \equiv X^n - 1 \pmod{n,\ X^r-1}$ forces n prime or $n^2 \equiv 1 \pmod r$; if true it yields a deterministic $\tilde{O}(\log^3 n)$ primality test. Lenstra and Pomerance gave a heuristic construction suggesting counterexamples exist but are astronomically large; none is known, and exhaustive search is hopeless. We pursue a *constructive* search for a counterexample of unrestricted size. Our contributions so far: (i) a reduction of the Lenstra–Pomerance construction to a two-sided subset-product problem over a partitioned smooth factor base; (ii) the observation that the candidate pool consists exactly of *twin smooth integers with prime sum* in a fixed congruence class, connecting the search to the isogeny-cryptography parameter literature and its datasets; (iii) feasibility estimates from that literature's empirical supply curves, isolating a single unmeasured quantity — the "coloring tax" — as the go/no-go criterion; and (iv) an obstruction: the boosting polynomials $2x^n - 1$ of the twin-smooth literature can never satisfy the local conditions of the Lenstra–Pomerance theorem as stated, motivating a re-derivation of its admissible local conditions.

1 Introduction

For coprime integers n, r write $T(-1, n, r)$ for the congruence

$$(X-1)^n \equiv X^n - 1 \pmod{n,\ X^r - 1}. \tag{1}$$

Conjecture 1.1 (Agrawal; Bhattacharjee–Pandey [2]). *If r is a prime not dividing n and $T(-1,n,r)$ holds, then n is prime or $n^2 \equiv 1 \pmod r$.*

If Conjecture 1.1 were true, a single evaluation of (1) at one $r = O(\log n)$ would certify primality, giving a deterministic $\tilde{O}(\log^3 n)$ test, versus $\tilde{O}(\log^6 n)$ for the best proven algorithm [7]. Its status is strange: Lenstra and Pomerance [3] gave a heuristic argument for $\gg x^{1-\varepsilon}$ counterexamples below x, yet the conjecture has survived exhaustive search to 10^{17} (Primaboinca, 2010–2020) and folklore places the least counterexample at hundreds of digits. Popovych [4] conjectured that adding the hypothesis $(X+2)^n \equiv X^n + 2 \pmod{n,\ X^r - 1}$ repairs the statement; his variant has no evidence against it.

Program. We abandon minimality. Any counterexample refutes Conjecture 1.1; the L–P density heuristic *improves* with size; and the construction of §2 is deterministic — when its arithmetic side conditions hold, (1) follows with no further luck required. The size of a counterexample produced this way is bounded only by the combinatorics of a subset-product problem (§3), whose raw material turns out to be the *twin smooth integers* of isogeny-based cryptography (§4). Verifying a candidate n of even 10^5 digits directly against (1) costs a few core-hours, and any find will also be tested against Popovych's second congruence.

2 The Lenstra–Pomerance construction

Theorem 2.1 (Lenstra–Pomerance [3]; statement following Váňa [5], who proves the $k \equiv 3 \pmod 4$ case)**.** *Let $n = p_1 \cdots p_k$ with $p_1, \ldots, p_k$ distinct primes such that*

(a) k is odd;

(b) $p_i \equiv 3 \pmod{80}$ for all i;

(c) $p_i - 1 \mid n - 1$ for all i *(n is Carmichael);*

(d) $p_i + 1 \mid n + 1$ for all i *(n is Lucas–Carmichael).*

Then $T(-1, n, 5)$ holds and $n^2 \not\equiv 1 \pmod 5$; in particular such an n with $k \geq 3$ is a counterexample to Conjecture 1.1.

Proof sketch. Since $p \equiv 3 \pmod 5$ has order 4 in $(\mathbb{Z}/5)^\times$, over $\mathbb{F}_p$ the polynomial $X^5 - 1$ factors as $(X-1)\Phi_5(X)$ with Φ_5 irreducible, so $\mathbb{Z}[X]/(p, X^5 - 1) \cong \mathbb{F}_p \times \mathbb{F}_{p^4}$. The $\mathbb{F}_p$ component of (1) is trivial. In $\mathbb{F}_{p^4}$, with ζ a primitive fifth root of unity, the Frobenius gives $(\zeta - 1)^{p^j} = \zeta^{p^j} - 1$ for every j. Conditions (c),(d) give $n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1}$, i.e. $n \equiv p \equiv p^3 \pmod{\operatorname{lcm}(p-1, p+1)}$, and the mod-80 condition resolves the 2-adic ambiguity; Váňa's analysis of the quantity $\rho(p) \mid 10(p^2-1)$ [5, Thm. 3.4] shows control modulo $10(p^2 - 1)$ suffices to conclude $(\zeta - 1)^n = \zeta^n - 1$ in each residue field. Lemma 2.2 gives $n^2 \not\equiv 1 \pmod 5$. **[TODO: Phase 0: full independent re-derivation, both k mod 4 cases; see Question 6.2.]** □

Lemma 2.2. *If $p_i \equiv 3 \pmod 5$ for all i and k is odd, then $n \equiv 3^k \equiv \pm 2 \pmod 5$; hence $n^2 \equiv 4 \not\equiv 1 \pmod 5$.*

Proof. $3^k \bmod 5$ cycles $3, 4, 2, 1$ with period 4; odd k gives 3 or 2, i.e. $\pm 2 \pmod 5$. □

The difficulty of satisfying (c) and (d) simultaneously is concentrated in the following trivial but decisive observation.

Lemma 2.3 (Side-coprimality)**.** *If n satisfies (c) and (d), then $\gcd(p_i - 1, p_j + 1) \mid 2$ for all i, j (including $i = j$). Consequently the odd primes dividing $\prod_i (p_i - 1)$ and those dividing $\prod_i (p_i + 1)$ form disjoint sets.*

Proof. An odd prime q dividing $p_i - 1$ and $p_j + 1$ divides both $n - 1$ and $n + 1$, hence their difference 2, a contradiction. □

Remark 2.4. No integer that is simultaneously Carmichael and Lucas–Carmichael is known; none exists below 10^{18} [8]. Lemma 2.3 is the structural reason, and the reason the *least* counterexample is expected to be enormous. It does not, however, obstruct *large* solutions, which is what we seek.

Lemma 2.5 (2-adic bookkeeping)**.** *If $p \equiv 3 \pmod{16}$ then $v_2(p-1) = 1$ and $v_2(p+1) = 2$; if $p \equiv 3 \pmod 5$ then $5 \nmid (p-1)(p+1)$. If moreover every $p_i \equiv 3 \pmod 4$ and k is odd, then $n \equiv 3 \pmod 4$, so the 2-power parts of conditions (c),(d) hold automatically.*

Proof. $p \equiv 3 \pmod{16}$ gives $p - 1 \equiv 2$ and $p + 1 \equiv 4 \pmod{16}$, fixing the stated 2-adic valuations; $p \equiv 3 \pmod 5$ gives $p \mp 1 \equiv 2, 4 \pmod 5$. For the last claim, $n \equiv 3^k \equiv 3 \pmod 4$ for odd k, so $2 \mid n - 1$ with $v_2(n-1) \geq 1 = v_2(p_i - 1)$ and $4 \mid n + 1$ with $v_2(n+1) \geq 2 = v_2(p_i + 1)$. □

Thus under condition (b) the entire problem lives in odd moduli prime to 5.

3 Reduction to a two-sided subset-product problem

Fix disjoint finite sets Q_-, Q_+ of odd primes with $5 \notin Q_- \cup Q_+$, and define the *pool*

$$\mathcal{P} \;=\; \Big\{ p \text{ prime} : p \equiv 3 \pmod{80},\ \tfrac{p-1}{2} \text{ is } Q_-\text{-smooth},\ \tfrac{p+1}{4} \text{ is } Q_+\text{-smooth} \Big\}.$$

Let $\Lambda_- = 2\,\mathrm{lcm}\{p-1 : p \in \mathcal{P}\}$ and $\Lambda_+ = 4\,\mathrm{lcm}\{p+1 : p \in \mathcal{P}\}$.

Proposition 3.1. *Let $S \subseteq \mathcal{P}$ with $|S|$ odd, $|S| \geq 3$, and $n = \prod_{p \in S} p$. If*

$$n \equiv 1 \pmod{\Lambda_-} \qquad \textit{and} \qquad n \equiv -1 \pmod{\Lambda_+}, \tag{2}$$

then n satisfies conditions (a)–(d) of Theorem 2.1, hence is a counterexample to Conjecture 1.1.

Proof. (a),(b) hold by construction of $\mathcal{P}$ and $|S|$. For (c): $p_i - 1 \mid \Lambda_-$ by definition of Λ_-, and $\Lambda_- \mid n-1$ by (2). Likewise (d) via $\Lambda_+ \mid n+1$. The elements of S are distinct primes, so n is squarefree with $k = |S| \geq 3$, hence composite. Note $\gcd(\Lambda_-, \Lambda_+) = 2$ by Lemma 2.3 (disjointness of $Q_\pm$) and Lemma 2.5, so (2) is consistent. □

Condition (2) is *stronger* than (c),(d) — it imposes the full lcm rather than the lcm over S — but in exchange it is linear: taking discrete logarithms in each cyclic component of $(\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times$ (Pohlig–Hellman [11] is cheap, all component orders being smooth by design), (2) becomes a 0/1 subset-sum over $\bigoplus_i \mathbb{Z}/m_i$, with one extra $\mathbb{Z}/2$ coordinate enforcing $|S|$ odd. The heuristic solvability condition is

$$2^{|\mathcal{P}|} \;\gg\; |G|, \qquad G = (\mathbb{Z}/\Lambda_-)^\times \times (\mathbb{Z}/\Lambda_+)^\times, \tag{3}$$

i.e. $|\mathcal{P}| \gtrsim \log_2 \Lambda_- + \log_2 \Lambda_+$ with slack for the solver. Candidate solvers, in escalating order: staged CRT in the style of Löh–Niebuhr's Carmichael factories [10], Wagner's k-list generalized birthday algorithm [12] (designed for exactly this pool-rich regime), and lattice reduction only for terminal low-dimensional corrections. **[TODO: Phase 2: work out the staged-CRT schedule and the expected $|S|$.]**

4 The twin-smooth dictionary

Lemma 4.1 (Dictionary). *Let $m \geq 1$ and $p = 2m+1$. Then $p \in \mathcal{P}$ (for $Q_\pm$ large enough to contain the relevant primes) if and only if*

(i) m and $m+1$ are both smooth with respect to $Q_- \cup Q_+ \cup \{2\}$, with the odd part of m being Q_--smooth and the odd part of $m+1$ being Q_+-smooth;

(ii) $m \equiv 1 \pmod{40}$;

(iii) p is prime.

Moreover (ii) alone implies $p \equiv 3 \pmod{80}$, $v_2(p-1) = 1$, $v_2(p+1) = 2$, and $5 \nmid m(m+1)$.

Proof. $p - 1 = 2m$ and $p + 1 = 2(m+1)$, so the smoothness conditions on $(p-1)/2 = m$ and $(p+1)/4 = (m+1)/2$ are exactly (i) once the 2-adic valuations match. For (ii): $m \equiv 1 \pmod{40} \iff p = 2m+1 \equiv 3 \pmod{80}$. Then m is odd (so $v_2(p-1) = 1$), $m \equiv 1 \pmod 4$ gives $m+1 \equiv 2 \pmod 4$ (so $v_2(p+1) = 2$), and $m \equiv 1 \pmod 5$ gives $m \not\equiv 0$, $m+1 \equiv 2 \not\equiv 0 \pmod 5$. □

Pairs of consecutive smooth integers $(m, m+1)$ — *twin smooths* — with $2m+1$ prime are precisely the parameters sought for the isogeny-based schemes B-SIDH [13] and SQIsign, and 2020–2025 produced an industrial literature on generating them:

- **PTE sieving** (Costello–Meyer–Naehrig [14]): Prouhet–Tarry–Escott solutions give degree-n polynomial pairs differing by a constant that split into linear factors; yielded 240–256-bit primes p with p^2-1 being 2^{15}-smooth, up to 512-bit at 2^{28}-smooth.
- **CHM** (Bruno et al. [15], optimizing Conrey–Holmström–McLaughlin [16]): iterative generation of *near-complete* sets of twin B-smooths. Headline data: $B = 547$ yields 82,026,426 pairs, the largest of 122 bits; implementation public.
- **Pell/Størmer** (Buzek–Hasan–Liu–Naehrig–Vigil [17]; classical [20, 21]): for fixed B the set of twin B-smooths is *finite*, computable via Pell equations $x^2 - 2Dy^2 = 1$; complete sets known provably for $B \le 100$ and heuristically for $B \le 113$ [19].
- **SVP records** (Mulder–Sterner–van Woerden [19]): shortest vectors in a prime-number lattice locate the *largest* twin for a given B (records: 196-bit at $B = 751$, 213-bit at $B = 997$), with an estimator for the largest twin as a function of B.
- **Boosting** (in [15, 18]): evaluate $p_n(x) = 2x^n - 1$ at twin-smooth arguments to reach cryptographic sizes.

By Lemma 4.1, a 1/40 congruence slice of *any* twin-smooth dataset, filtered for primality of $2m+1$, is a ready-made pool $\mathcal{P}$ — before the one condition their literature does not consider: the global side-partition of Lemma 2.3.

5 Feasibility estimates

5.1 Supply and demand at $B = 547$

Demand. $\log_2 \Lambda_- + \log_2 \Lambda_+ \le \sum_{2<q\le B} \log_2 q^{e_q}$ where q^{e_q} is the largest power appearing in any used $p \mp 1$; the side-partition splits this sum but does not double it. With $\pi(547) = 101$ primes and exponents capped by observed usage, this is $\approx \theta(547)/\ln 2 \approx 750$ bits plus modest slack from powers of small primes: call it 1000–1500 bits, and less if the chosen subpool touches only a sub-universe of primes.

Supply. 82M pairs at $B = 547$, $\div 40$ for the congruence slice (Lemma 4.1(ii)), $\div \ln p \approx 20$–55 for primality of $2m+1$ at the dataset's typical sizes: roughly 5×10^4 pool candidates *from existing data*. **[TODO: Phase 1a: regenerate the dataset and replace these estimates with exact histograms.]**

The cushion between supply and demand is therefore a factor ≈ 25–40. Moreover it *grows* with B: demand grows linearly ($\approx 1.44\,B$ bits) while empirical twin counts grow like B^α with $\alpha \approx 2.5$–3 (from $B = 113 \to 547$). Størmer finiteness caps the supply at each fixed B, but the ceiling recedes superlinearly.

5.2 The coloring tax

The unmeasured factor is the side-partition. Each pool element consumes $\omega_-(p)+\omega_+(p)$ prime-side assignments (the number of distinct odd primes dividing $(p-1)/2$ and $(p+1)/4$), and all chosen elements must respect one partition $Q_- \sqcup Q_+$. Under a uniformly random balanced partition, an element survives with probability $\approx 2^{-(\omega_-+\omega_+)}$, so the expected number of survivors is $\sum_{p\in\mathcal{P}} 2^{-\omega(p)}$.

At $B = 547$, bulk-of-distribution twins (30–50 bits) have $\omega_- + \omega_+ \approx 8$–14, predicting only *tens* of survivors from 5×10^4 candidates — two orders of magnitude short of the $\sim$1500–3000 needed.

The program therefore rests on the question:

Question 5.1. *By how much does an* optimized *partition beat a random one? Survivors correlate (elements sharing factor support survive together), supply is thickest exactly where ω is smallest, and demand shrinks with the effective prime universe. The complete Størmer sets ($B \le 113$) permit computing the exact optimum at small scale; the CHM set at $B = 547$ permits evaluating it at target scale.* ***[TODO: Phase 1a go/no-go: survival rate $\gtrsim 5\%$ means the direct route is live; $\lesssim 0.1\%$ kills it at fixed B.]***

5.3 Heuristic assumptions ledger

Statement	Status	Role
$\gcd(p-1, p+1) = 2$; Lemmas 2.2–4.1	theorems	structure
Størmer finiteness at fixed B	theorem	supply ceiling
Local densities $P(q \mid p \mp 1) = \frac{1}{q-1}$, exclusive	exact	constants only
Two-sided smoothness independence ($\rho(u_-)\rho(u_+)$)	conjectural	steering only
One-sided density $\#\{p : p-1 \text{ smooth}\} \sim \rho(u)\pi(x)$	open (Erdős–Pomerance line)	steering only
Empirical supply curves (CHM, Størmer sets)	data	replaces both

6 An obstruction to boosting, and admissible local conditions

The boosting polynomials $p_n(x) = 2x^n - 1$ of [15, 18] are structurally attractive here: for $n = 2$, $(p+1)/2 = x^2$ and $(p-1)/2 = (x-1)(x+1)$ with $\gcd(x, x^2 - 1) = 1$, so the two sides have automatically disjoint prime support — Lemma 2.3 satisfied by construction — and the two-sided smoothness burden moves to $(x, x^2 - 1)$ at *half* the bit-size, where Dickman densities are far kinder. Unfortunately:

Proposition 6.1. *For every $n \ge 2$ and every integer x: $2x^n - 1 \not\equiv 3 \pmod{16}$. In particular no value of any p_n, $n \ge 2$, satisfies condition (b) of Theorem 2.1. For $n = 2$ the obstruction is doubled: $x^2 \equiv 2 \pmod 5$ is also unsolvable, while for $n = 3$ the mod-5 condition is satisfiable ($x \equiv 3 \pmod 5$) and only the 2-adic obstruction remains.*

Proof. $2x^n \equiv 4 \pmod{16}$ requires $x^n \equiv 2 \pmod 8$. If x is even then $4 \mid x^n$ (as $n \ge 2$), so $x^n \equiv 0$ or $4 \pmod 8$; if x is odd then x^n is odd. In no case is $x^n \equiv 2 \pmod 8$. For the mod-5 claims: the squares modulo 5 are $\{0, 1, 4\} \not\ni 2$, while $3^3 = 27 \equiv 2 \pmod 5$. □

The obstruction is to condition (b) *as stated*, not (necessarily) to the theorem's mechanism. Condition (b) fixes one convenient cell of local conditions ($p \equiv 3 \bmod 16$ giving $v_2(p \mp 1) = (1, 2)$, and $\mathrm{ord}_5(p) = 4$); Váňa's $\rho(p)$ framework suggests other cells may be admissible, e.g. pools with a uniform but different 2-adic pattern $v_2(p+1) = v$ for $v > 2$, compensated in the mod-2^k component of the target congruence (2).

Question 6.2 (Admissible local cells)**.** *Determine the full set of local conditions (residues mod 2^k and mod 5, allowed v_2 patterns, and the required uniformity across the pool) under which the conclusion of Theorem 2.1 survives. In particular: is any admissible cell reachable by $p_n(x) = 2x^n - 1$ for some n, or by another side-separating polynomial family? A positive answer would collapse the coloring tax of §5.2 structurally and halve the effective smoothness size.*

For $n = 3$: $p = 2x^3 - 1$ has $v_2(p + 1) = 1 + 3v_2(x)$, so $x \equiv 2 \pmod 4$ gives the uniform pattern $v_2(p + 1) = 4$, $v_2(p - 1) = 1$ — a natural first test case for Question 6.2. **[TODO: Phase 0.]**

7 Program

P0. Independent re-derivation of Theorem 2.1; resolution of Question 6.2; numerical unit tests of the congruence machinery; prior-art due diligence (in particular whether the L–P construction was ever run at scale).

P1a. Data-first go/no-go: regenerate the CHM $B = 547$ set, apply the dictionary filters, and answer Question 5.1 by direct optimization; calibrate against the complete $B \le 113$ sets.

P1b. If needed, a native harvester (constructive enumeration of one side, trial division on the other) at larger B or targeted congruences.

P2. Subset-product solver (staged CRT, Wagner), with the parity coordinate.

P3. Assembly and dual independent verification of any find, including the direct computation of $T(-1, n, 5)$ and the Popovych congruence; write-up (a quantified negative result at fixed B is also publishable).

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